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RESULTS

Results

01 · EXPLORATORY FACTOR ANALYSIS (EFA)

discover underlying factors

Table 1. Suitability

KMO	Bartlett's χ^2	df	p
.75	904.1	36	<.001

Table 2. Variance explained

Factor	Eigenvalue	% variance	Cumulative %
F1	2.24	24.9	24.9
F2	1.34	14.9	39.8
F3	1.28	14.2	54.0

Table 3. Rotated factor loadings

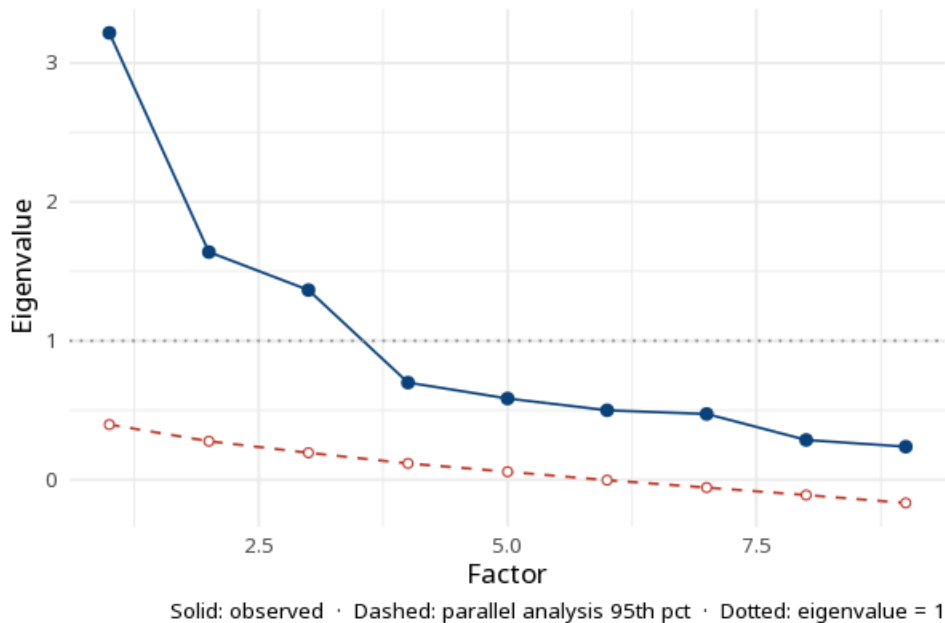
Item	F1	F2	F3	Communality
x1		.59		.48
x2		.51		.26
x3		.69		.45
x4	.85			.73
x5	.89			.75
x6	.81			.69
x7			.73	.51
x8			.69	.52
x9		.38	.46	.46

Table 4. Interfactor correlations (Φ)

	F1	F2	F3
F1	1.00		
F2	0.32	1.00	
F3	0.21	0.26	1.00

Loading columns expand to the number of retained factors; loadings $< .32$ suppressed (Tabachnick & Fidell). Φ (interfactor correlations) shown for oblique rotation (oblimin) only — omitted for varimax (orthogonal). KMO $\geq .60$ acceptable, $\geq .70$ preferred (Kaiser & Rice, 1974). Bartlett $p < .001$ required. Parallel analysis (Horn, 1965) is the preferred retention rule; Kaiser eigenvalue > 1 is offered but systematically over-extracts (Zwick & Velicer, 1986). Factor order: descending SS-loadings.

Figure. Factor retention



Understanding the numbers

KMO. Sampling adequacy - whether the variables share enough variance to factor well; $> .60$ acceptable, $> .70$ preferred (Kaiser & Rice, 1974). Here, KMO = .75.

Bartlett's χ^2 . Tests whether the correlation matrix differs from an identity matrix (no correlation at all) - a prerequisite check for factoring. Here, Bartlett's $\chi^2(36) = 904.1$.

df. Degrees of freedom for Bartlett's test, based on the number of items. Here, $df = 36$.

p. The significance of Bartlett's test; $p < .001$ is required for suitability. Here, $p < .001$.

Factor. One retained factor from the chosen retention rule - parallel analysis (Horn, 1965) is recommended over the Kaiser eigenvalue > 1 rule, which over-extracts (Zwick & Velicer, 1986). Here, 3 factor(s) were retained (rows below), ordered by descending variance explained.

Eigenvalue. How much total variance a factor accounts for; parallel analysis retains a factor only once its eigenvalue clears a randomly-generated threshold. Each retained factor's eigenvalue already cleared the retention rule you selected.

% variance. The share of total item variance one factor explains on its own. Each row's % variance is that factor's own individual contribution.

Cumulative %. The running total of variance explained as factors accumulate down the table. Here, the retained factors together explain 54.0% of total variance.

Factor 1 loading. Each item's unique contribution to Factor 1; loadings $|\lt .32|$ are suppressed, and a loading $\geq .32$ on two or more factors flags a cross-loader (Tabachnick & Fidell). A blank cell here means that item's loading on Factor 1 fell below the $|\lt .32|$ display threshold.

Factor 2 loading. Each item's unique contribution to Factor 2, on the same $|\lt .32|$ suppression rule as Factor 1 (Tabachnick & Fidell). A blank cell here means that item's loading on Factor 2 fell below the $|\lt .32|$ display threshold.

Communality. The share of an item's variance explained by all retained factors together. Each item's communality summarizes how well the retained factors, together, account for it.

How to read this test

Check KMO ($> .60$ acceptable, $> .70$ preferred; Kaiser & Rice, 1974) and a significant Bartlett's test ($p < .001$) for suitability. Parallel analysis (Horn, 1965) is the recommended retention rule — it corrects the upward bias of the Kaiser eigenvalue > 1 rule (Zwick & Velicer, 1986). Table 3 shows the rotated pattern matrix: each loading is the unique contribution of the factor to that item. Loadings $|\lt .32|$ ($\sim 10\%$ variance) are suppressed; a loading $\geq .32$ on two or more factors flags a cross-loader (Tabachnick & Fidell). For oblique rotation (oblimin), the Φ matrix shows interfactor correlations — if all $|\Phi| < .32$, orthogonal rotation (varimax) may be preferred. EFA is exploratory — it discovers structure; to confirm a proposed factor structure run a CFA, ideally on a separate sample (Watkins, 2018).

APA template: EFA (KMO = .75, Bartlett's $\chi^2(36) = 904.1$, $p < .001$) with parallel analysis retained 3 factors explaining 54.0% of variance (oblimin rotation).

Statistical basis: KMO sampling-adequacy verbal labels ($\geq .60$ acceptable, $\geq .70$ preferred) (Kaiser, H. F., Rice, J. (1974). "Little Jiffy, Mark IV." Educational and Psychological Measurement, 34(1), 111-117.); Parallel analysis as the preferred factor-retention rule (Horn, J. L. (1965). "A rationale and test for the number of factors in factor analysis." Psychometrika, 30(2), 179-185.); Kaiser eigenvalue > 1 rule systematically over-extracts (Zwick, W. R., Velicer, W. F. (1986). "Comparison of five rules for determining the number of components to retain." Psychological Bulletin, 99(3), 432-442.); EFA best-practice reporting standard (Watkins, M. W. (2018). "Exploratory Factor Analysis: A Guide to Best Practice." Journal of Black Psychology, 44(3), 219-246.). Full references in the exported CITATIONS.txt.

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